EXAMPLE 1 Evaluate $\int_{-1}^{1} 3x^2\sqrt{x^3 + 1}\,dx$.

EXAMPLE 7 Evaluate $\int_{0}^{4} \sqrt{2x + 1}\,dx$ using (6).

EXAMPLE 8 Evaluate $\int_{1}^{2} \frac{dx}{(3 - 5x)^2}$.

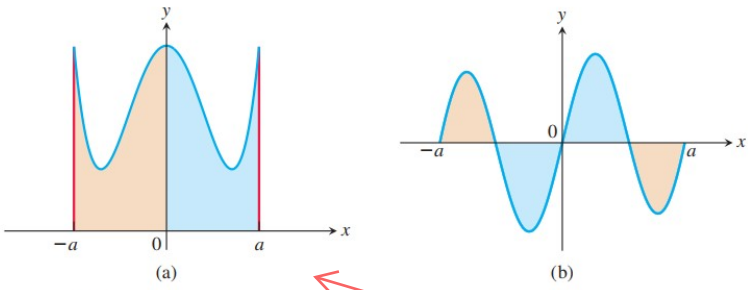
EXAMPLE 9 Calculate $\int_{1}^{e} \frac{\ln x}{x}\,dx$.

(a) $\int_{\pi/4}^{\pi/2} \cot\theta \csc^2\theta\,d\theta =$.

(b) $\int_{0}^{\pi/2} \frac{2\sin x \cos x}{(1 + \sin^2 x)^3}\,dx =$.

Definite Integrals of Symmetric Functions

The Substitution Formula in Theorem 7 simplifies the calculation of definite integrals of even and odd functions (Section 1.1) over a symmetric interval $[-a, a]$ (Figure 5.24).



odd func.
symetrical to origin

even func.
symetrical to y-axis

FIGURE 5.24 (a) For f an even function, the integral from $-a$ to a is twice the integral from 0 to a. (b) For f an odd function, the integral from $-a$ to a equals 0.

THEOREM 8 Let f be continuous on the symmetric interval $[-a, a]$.

(a) If f is even, then $\int_{-a}^{a} f(x)\,dx = 2\int_{0}^{a} f(x)\,dx.$

(b) If f is odd, then $\int_{-a}^{a} f(x)\,dx = 0.$

EXAMPLE 3 Evaluate $\int_{-2}^{2} (x^4 - 4x^2 + 6)\,dx$.

EXAMPLE 10 Since $f(x) = x^6 + 1$ satisfies $f(-x) = f(x)$, it is even and so

EXAMPLE 11 Since $f(x) = (\tan x)/(1 + x^2 + x^4)$ satisfies $f(-x) = -f(x)$, it is odd and so

5.5 EXERCISES

1–6 Evaluate the integral by making the given substitution.

1. $\int e^{-x}\,dx, \quad u = -x$
2. $\int x^3(2 + x^4)^5\,dx, \quad u = 2 + x^4$
3. $\int x^2\sqrt{x^3 + 1}\,dx, \quad u = x^3 + 1$
4. $\int \frac{dt}{(1 - 6t)^4}, \quad u = 1 - 6t$
5. $\int \cos^3\theta \sin\theta\,d\theta, \quad u = \cos\theta$
6. $\int \frac{\sec^2(1/x)}{x^2}\,dx, \quad u = 1/x$

7–46 Evaluate the indefinite integral.

7. $\int x \sin(x^2)\,dx$
8. $\int x^2(x^3 + 5)^9\,dx$
9. $\int (3x - 2)^{20}\,dx$
10. $\int (3t + 2)^{2.4}\,dt$
11. $\int (x + 1)\sqrt{2x + x^2}\,dx$
12. $\int \frac{x}{(x^2 + 1)^2}\,dx$
13. $\int \frac{dx}{5 - 3x}$
14. $\int e^x \sin(e^x)\,dx$
15. $\int \sin \pi t\,dt$
16. $\int \frac{x}{x^2 + 1}\,dx$
17. $\int \frac{a + bx^2}{\sqrt{3ax + bx^3}}\,dx$
18. $\int \sec 2\theta \tan 2\theta\,d\theta$

19. $\int \frac{(\ln x)^2}{x}\,dx$

20. $\int \frac{dx}{ax+b} \quad (a \neq 0)$

21. $\int \frac{\cos\sqrt{t}}{\sqrt{t}}\,dt$

22. $\int \sqrt{x}\,\sin(1+x^{3/2})\,dx$

23. $\int \cos\theta\,\sin^6\theta\,d\theta$

24. $\int (1+\tan\theta)^5\sec^2\theta\,d\theta$

25. $\int e^x\sqrt{1+e^x}\,dx$

26. $\int e^{\cos t}\sin t\,dt$

27. $\int \frac{z^2}{\sqrt[3]{1+z^3}}\,dz$

28. $\int \frac{\tan^{-1}x}{1+x^2}\,dx$

29. $\int e^{\tan x}\sec^2 x\,dx$

30. $\int \frac{\sin(\ln x)}{x}\,dx$

31. $\int \frac{\cos x}{\sin^2 x}\,dx$

32. $\int \frac{e^x}{e^x+1}\,dx$

33. $\int \sqrt{\cot x}\,\csc^2 x\,dx$

34. $\int \frac{\cos(\pi/x)}{x^2}\,dx$

55. $\int_0^{\pi} \sec^2(t/4)\,dt$

56. $\int_{1/6}^{1/2} \csc \pi t\,\cot \pi t\,dt$

57. $\int_{-\pi/6}^{\pi/6} \tan^3\theta\,d\theta$

58. $\int_0^1 xe^{-x^2}\,dx$

59. $\int_1^2 \frac{e^{1/x}}{x^2}\,dx$

60. $\int_{-\pi/2}^{\pi/2} \frac{x^2\sin x}{1+x^6}\,dx$

61. $\int_0^{13} \frac{dx}{\sqrt[3]{(1+2x)^2}}$

62. $\int_0^{\pi/2} \cos x\,\sin(\sin x)\,dx$

63. $\int_0^a x\sqrt{x^2+a^2}\,dx \quad (a>0)$

64. $\int_0^a x\sqrt{a^2-x^2}\,dx$

65. $\int_1^2 x\sqrt{x-1}\,dx$

66. $\int_0^4 \frac{x}{\sqrt{1+2x}}\,dx$

67. $\int_e^{e^4} \frac{dx}{x\sqrt{\ln x}}$

68. $\int_0^{1/2} \frac{\sin^{-1}x}{\sqrt{1-x^2}}\,dx$

69. $\int_0^1 \frac{e^z+1}{e^z+z}\,dz$

70. $\int_0^{T/2} \sin(2\pi t/T-\alpha)\,dt$

Areas Between Curves

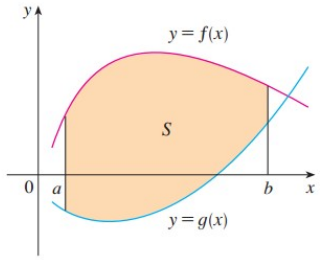
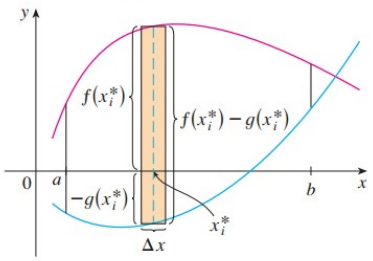


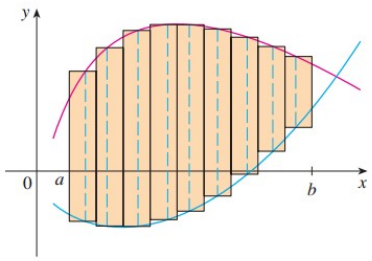
FIGURE 1
$S = \{(x, y) \mid a \leq x \leq b, g(x) \leq y \leq f(x)\}$

$$\sum_{i=1}^{n} [f(x_i^*) - t(x_i^*)]\,\Delta x$$

is therefore an approximation to what we intuitively think of as the area of S.



(a) Typical rectangle



(b) Approximating rectangles

$$A = \lim_{n \to \infty} \sum_{i=1}^{n} [f(x_i^*) - t(x_i^*)]\,\Delta x$$

2 The area A of the region bounded by the curves $y = f(x)$, $y = g(x)$, and the lines $x = a$, $x = b$, where f and t are continuous and $f(x) \geq t(x)$ for all x in $[a, b]$, is

$$A = \int_a^b [f(x) - g(x)]\,dx$$

DEFINITION If f and g are continuous with $f(x) \geq g(x)$ throughout $[a, b]$, then the **area of the region between the curves $y = f(x)$ and $y = g(x)$ from a to b** is the integral of $(f - g)$ from a to b:

$$A = \int_a^b [f(x) - g(x)]\, dx.$$

EXAMPLE 4 Find the area of the region enclosed by the parabola $y = 2 - x^2$ and the line $y = -x$.

EXAMPLE 1 Find the area of the region bounded above by $y = e^x$, bounded below by $y = x$, and bounded on the sides by $x = 0$ and $x = 1$.

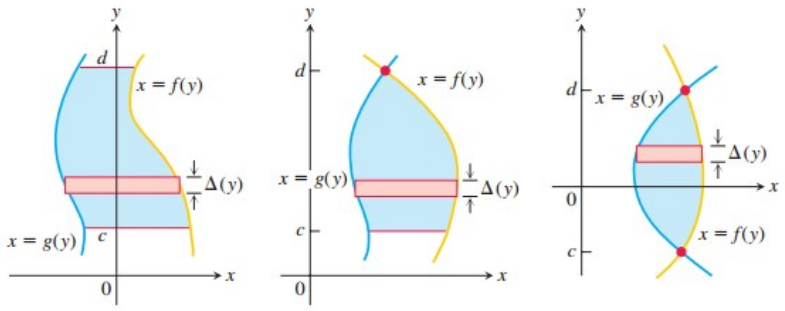
EXAMPLE 2 Find the area of the region enclosed by the parabolas $y = x^2$ and $y = 2x - x^2$.

EXAMPLE 5 Find the area of the region in the first quadrant that is bounded above by $y = \sqrt{x}$ and below by the x-axis and the line $y = x - 2$.

Integration with Respect to y

If a region's bounding curves are described by functions of y, the approximating rectangles are horizontal instead of vertical and the basic formula has y in place of x.

For regions like these:



use the formula

$$A = \int_c^d [\, f(y) - g(y) \,]\, dy.$$

In this equation f always denotes the right-hand curve and g the left-hand curve, so $f(y) - g(y)$ is nonnegative.

EXAMPLE 5 Find the area of the region in the first quadrant that is bounded above by $y = \sqrt{x}$ and below by the x-axis and the line $y = x - 2$.

EXAMPLE 6 Find the area enclosed by the line $y = x - 1$ and the parabola $y^2 = 2x + 6$.

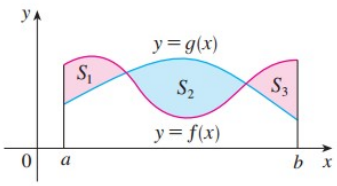


FIGURE 9

If we are asked to find the area between the curves $y = f(x)$ and $y = t(x)$ where $f(x) \geqslant t(x)$ for some values of x but $t(x) \geqslant f(x)$ for other values of x, then we split the given region S into several regions $S_1, S_2, \ldots$ with areas $A_1, A_2, \ldots$ as shown in Figure 9. We then define the area of the region S to be the sum of the areas of the smaller regions $S_1, S_2, \ldots$, that is, $A = A_1 + A_2 + \cdots$. Since

$$|f(x) - g(x)| = \begin{cases} f(x) - g(x) & \text{when } f(x) \geqslant g(x) \\ g(x) - f(x) & \text{when } g(x) \geqslant f(x) \end{cases}$$

we have the following expression for A.

3 The area between the curves $y = f(x)$ and $y = g(x)$ and between $x = a$ and $x = b$ is

$$A = \int_a^b |f(x) - g(x)|\, dx$$

When evaluating the integral in (3), however, we must still split it into integrals corresponding to $A_1, A_2, \ldots$.

EXAMPLE 5 Find the area of the region bounded by the curves $y = \sin x$, $y = \cos x$, $x = 0$, and $x = \pi/2$.

EXAMPLE 7 Find the area of the region bounded below by the line $y = 2 - x$ and above by the curve $y = \sqrt{2x - x^2}$.

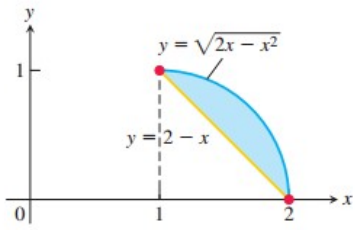
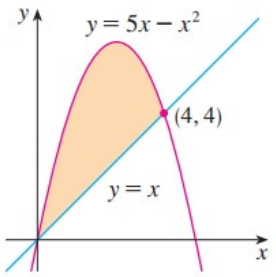


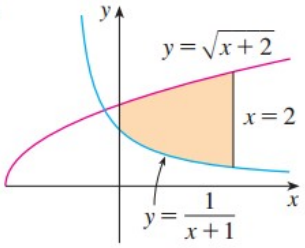
FIGURE 5.32 The region described by the curves in Example 7.

1–4 Find the area of the shaded region.

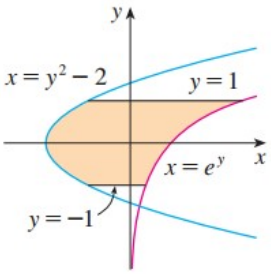
1.



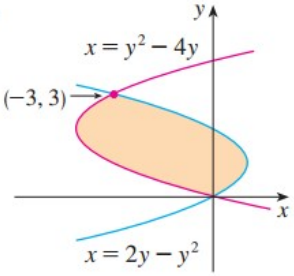
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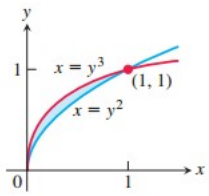
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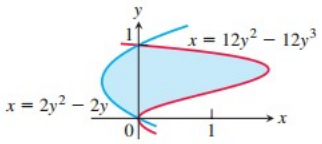
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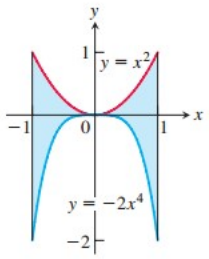
32.



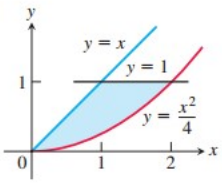
33.



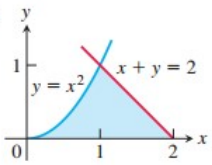
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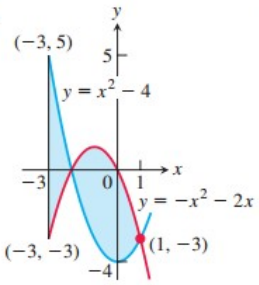
35.



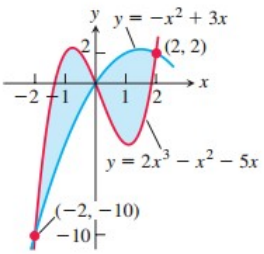
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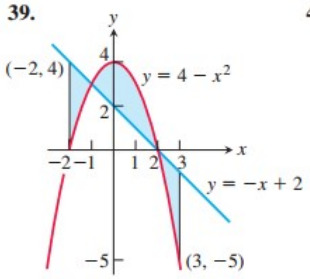
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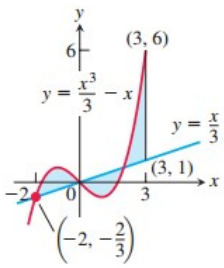
38.



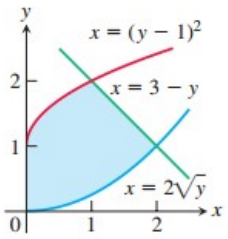
39.



40.



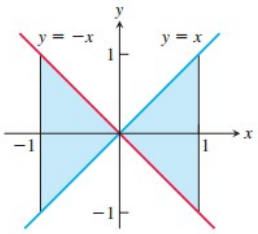
78. Find the area of the region in the first quadrant bounded on the left by the y-axis, below by the curve $x = 2\sqrt{y}$, above left by the curve $x = (y - 1)^2$, and above right by the line $x = 3 - y$.



81. Which of the following integrals, if either, calculates the area of the shaded region shown here? Give reasons for your answer.

a. $\int_{-1}^{1} (x - (-x))\,dx = \int_{-1}^{1} 2x\,dx$

b. $\int_{-1}^{1} (-x - (x))\,dx = \int_{-1}^{1} -2x\,dx$



5–28 Sketch the region enclosed by the given curves. Decide whether to integrate with respect to x or y. Draw a typical approximating rectangle and label its height and width. Then find the area of the region.

5. $y = x + 1, \quad y = 9 - x^2, \quad x = -1, \quad x = 2$

6. $y = \sin x, \quad y = e^x, \quad x = 0, \quad x = \pi/2$

7. $y = x, \quad y = x^2$

8. $y = x^2 - 2x, \quad y = x + 4$

9. $y = 1/x, \quad y = 1/x^2, \quad x = 2$

10. $y = 1 + \sqrt{x}, \quad y = (3 + x)/3$

11. $y = x^2, \quad y^2 = x$

12. $y = x^2, \quad y = 4x - x^2$

13. $y = 12 - x^2, \quad y = x^2 - 6$

14. $y = \cos x, \quad y = 2 - \cos x, \quad 0 \leq x \leq 2\pi$

15. $y = \tan x, \quad y = 2 \sin x, \quad -\pi/3 \leq x \leq \pi/3$

16. $y = x^3 - x, \quad y = 3x$

29–30 Use calculus to find the area of the triangle with the given vertices.

29. $(0, 0), \quad (2, 1), \quad (-1, 6)$

30. $(0, 5), \quad (2, -2), \quad (5, 1)$

31–32 Evaluate the integral and interpret it as the area of a region. Sketch the region.

31. $\int_0^{\pi/2} |\sin x - \cos 2x| \, dx$

32. $\int_0^4 |\sqrt{x+2} - x| \, dx$

33–34 Use the Midpoint Rule with $n = 4$ to approximate the area of the region bounded by the given curves.

33. $y = \sin^2(\pi x/4), \quad y = \cos^2(\pi x/4), \quad 0 \leq x \leq 1$

34. $y = \sqrt[3]{16 - x^3}, \quad y = x, \quad x = 0$

35–38 Use a graph to find approximate x-coordinates of the points of intersection of the given curves. Then find (approximately) the area of the region bounded by the curves.

35. $y = x \sin(x^2), \quad y = x^4$

36. $y = e^x, \quad y = 2 - x^2$

37. $y = 3x^2 - 2x, \quad y = x^3 - 3x + 4$

38. $y = x \cos x, \quad y = x^{10}$

6.1 Volumes Using Cross-Sections

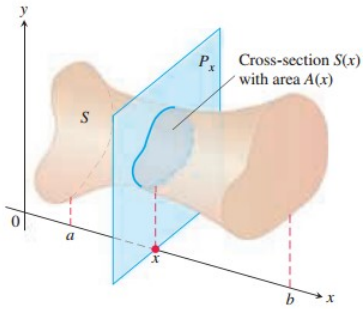


FIGURE 6.1 A cross-section $S(x)$ of the solid S formed by intersecting S with a plane P_x perpendicular to the x-axis through the point x in the interval $[a, b]$.



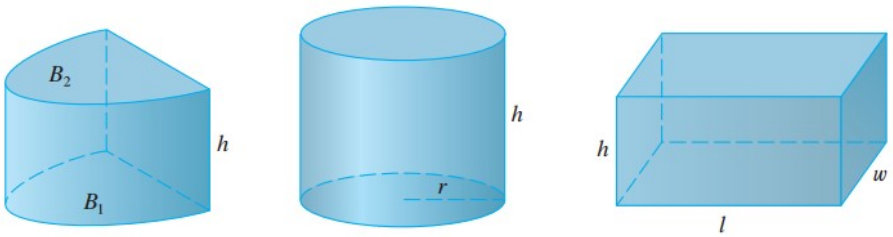
Plane region whose area we know

Cylindrical solid based on region
Volume = base area × height = Ah

FIGURE 6.2 The volume of a cylindrical solid is always defined to be its base area times its height.

$$V = Ah$$

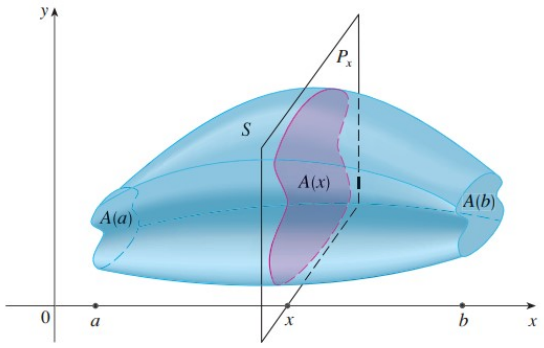
In particular, if the base is a circle with radius r, then the cylinder is a circular cylinder with volume $V = \pi r^2 h$ [see Figure 1(b)], and if the base is a rectangle with length l and width w, then the cylinder is a rectangular box (also called a *rectangular parallelepiped*) with volume $V = lwh$ [see Figure 1(c)].

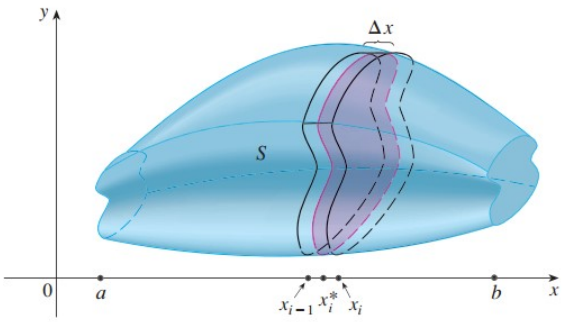


(a) Cylinder $V = Ah$

(b) Circular cylinder $V = \pi r^2 h$

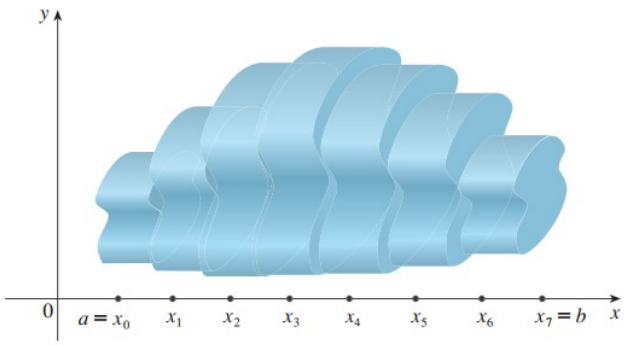
(c) Rectangular box $V = lwh$





The volume of this cylinder is $A(x_i^*)\,\Delta x$, so an approximation to our intuitive conception of the volume of the ith slab S_i is

$$V(S_i) \approx A(x_i^*)\,\Delta x$$



$$V \approx \sum_{i=1}^{n} A(x_i^*)\,\Delta x$$

This approximation appears to become better and better as $n \to \infty$.

DEFINITION OF VOLUME Let S be a solid that lies between $x = a$ and $x = b$. If the cross-sectional area of S in the plane P_x, through x and perpendicular to the x-axis, is $A(x)$, where A is a continuous function, then the **volume** of S is

$$V = \lim_{n \to \infty} \sum_{i=1}^{n} A(x_i^*)\,\Delta x = \int_a^b A(x)\,dx$$

Volume of the solid is integral of areas.

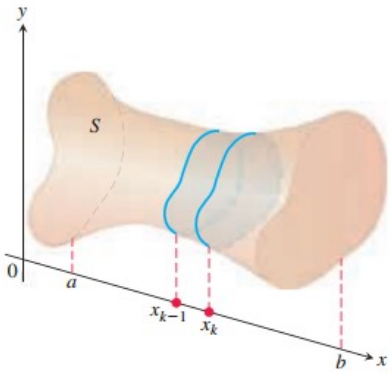


FIGURE 6.3 A typical thin slab in the solid S.

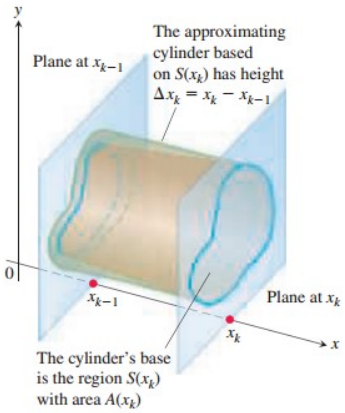


FIGURE 6.4 The solid thin slab in Figure 6.3 is shown enlarged here. It is approximated by the cylindrical solid with base $S(x_k)$ having area $A(x_k)$ and height $\Delta x_k = x_k - x_{k-1}$.

$$\text{Volume} = \text{area} \times \text{height} = A \cdot h.$$

In the method of slicing, the base will be the cross-section of S that has area $A(x)$, and the height will correspond to the width Δx_k of subintervals formed by partitioning the interval $[a, b]$ into finitely many subintervals $[x_{k-1}, x_k]$.

Slicing by Parallel Planes

We partition $[a, b]$ into subintervals of width (length) Δx_k and slice the solid, as we would a loaf of bread, by planes perpendicular to the x-axis at the partition points $a = x_0 < x_1 < \cdots < x_n = b$. These planes slice S into thin "slabs" (like thin slices of a loaf of bread). A typical slab is shown in Figure 6.3. We approximate the slab between the plane at x_{k-1} and the plane at x_k by a cylindrical solid with base area $A(x_k)$ and height $\Delta x_k = x_k - x_{k-1}$ (Figure 6.4). The volume V_k of this cylindrical solid is $A(x_k) \cdot \Delta x_k$, which is approximately the same volume as that of the slab:

$$\text{Volume of the } k\text{th slab} \approx V_k = A(x_k)\,\Delta x_k.$$

The volume V of the entire solid S is therefore approximated by the sum of these cylindrical volumes,

$$V \approx \sum_{k=1}^{n} V_k = \sum_{k=1}^{n} A(x_k)\,\Delta x_k.$$

This is a Riemann sum for the function $A(x)$ on $[a, b]$. The approximation given by this Riemann sum converges to the definite integral of $A(x)$ as $n \to \infty$:

$$\lim_{n\to\infty} \sum_{k=1}^{n} A(x_k)\,\Delta x_k = \int_a^b A(x)\,dx.$$

Therefore, we define this definite integral to be the volume of the solid S.

DEFINITION The **volume** of a solid of integrable cross-sectional area $A(x)$ from $x = a$ to $x = b$ is the integral of A from a to b,

$$V = \int_a^b A(x)\,dx.$$

